MEDIUM

ADVANCED MATH

Advanced Math

15

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

$$5x + 42x - 5 = 0$$

Which of the following is a solution to the given equation?

A. $-\frac{5}{2}$

B. $-\frac{5}{4}$

C. $-\frac{4}{5}$

D. $-\frac{2}{5}$

Q2

ADVANCED MATH

The height, in feet, of an object x seconds after it is thrown straight up in the air can be modeled by the function $h(x) = -16x^2 + 20x + 5$. Based on the model, which of the following statements best interprets the equation $h(1.4) = 1.64$?

A. The height of the object 1.4 seconds after being thrown straight up in the air is 1.64 feet.

B. The height of the object 1.64 seconds after being thrown straight up in the air is 1.4 feet.

C. The height of the object 1.64 seconds after being thrown straight up in the air is approximately 1.4 times as great as its initial height.

D. The speed of the object 1.4 seconds after being thrown straight up in the air is approximately 1.64 feet per second.

Q3

GRID-IN ADVANCED MATH

$$38x^2 = 389$$

What is the negative solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 GRID-IN ADVANCED MATH

What is an x-coordinate of an x-intercept of the graph of $y = 3x - 14x + 5x + 4$ in the xy-plane?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN ADVANCED MATH

If $4x - 4 = 112$, what is the positive value of $x - 1$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Time (years)	Total amount (dollars)
0	604.00
1	606.42
2	608.84

Rosa opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Rosa opened the account and the total amount n, in dollars, in the account. If Rosa made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n?

A. $n = 1 + 604^t$

B. $n = 1 + 0.004^t$

C. $n = 6041 + 0.004^t$

D. $n = 0.0041 + 604^t$

Q7

GRID-IN ADVANCED MATH

$$d - 30d + 30 - 7 = -7$$

What is a solution to the given equation?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

ADVANCED MATH

Bacteria are growing in a liquid growth medium. There were 300,000 cells per milliliter during an initial observation. The number of cells per milliliter doubles every 3 hours. How many cells per milliliter will there be 15 hours after the initial observation?

A. 1,500,000

B. 2,400,000

C. 4,500,000

D. 9,600,000

$$y = x^2 - 1$$
$$y = 3$$

When the equations above are graphed in the xy-plane, what are the coordinates (x, y) of the points of intersection of the two graphs?

A. $(2,3)$ and $(-2,3)$

B. $(2,4)$ and $(-2,4)$

C. $(3,8)$ and $(-3,8)$

D. $(\sqrt{2},3)$ and $(-\sqrt{2},3)$

Time (years)	Total amount (dollars)
0	670.00
1	674.02
2	678.06

Sara opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Sara opened the account and the total amount d, in dollars, in the account. If Sara made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and d?

A. $d = 0.0061 + 670^t$

B. $d = 6701 + 0.006^t$

C. $d = 1 + 0.006^t$

D. $d = 1 + 670^t$